Camera_Calibration

1. Calibration is the process to determine the intrinsic (K plus lens distortion) and extrinsic (R, T) parameters of a camera. For now, we will neglect the lens distortion and see later how it can be determined.
2. The solution for K, R, T can be found be applying the perspective projection equation:
    1. $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K\,[R|T] \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$
3. There are two popular methods:
    1. Tsai's method: uses 3D objects
    2. Zhang's method: uses planar grids
4. Tsai's method: calibration from 3D objects
    1. This method was proposed in 1987 by Tsai and consists of measuring the 3D position of $n \geq 6$ control points on a 3D calibration target and the 2D coordinates of their projection in the image.
    2. Assumption: we know 2D and 3D coordinates of control points
        1. 2D: image pre-processing (e.g. corner detection).
        2. 3D: we know actual size of the 3D object and actual positions of control points as well.
    3. Applying the Direct Linear Transform (DLT)
    4. The idea of the DLT is to rewrite the perspective projection equation as a **homogeneous linear equation** and solve it by standard methods. Let's write the perspective equation for a generic 3D-2D point correspondence:
    5. $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_u & 0 & u_0 \\ 0 & \alpha_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$
    6. $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$
    7. $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = M \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$
    8. $\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} m_1^T \\ m_2^T \\ m_3^T \end{bmatrix} \begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix}$, where m_i^T is the i-th row of M.
    9. let $\begin{bmatrix} X_W \\ Y_W \\ Z_W \\ 1 \end{bmatrix} = P$, then conversion back from homogeneous coordinates to pixel coordinates leads to:
    $u = \frac{\lambda u}{\lambda} = \frac{m_1^T \cdot P}{m_3^T \cdot P}, v = \frac{\lambda u}{\lambda} = \frac{m_2^T \cdot P}{m_3^T \cdot P} \Rightarrow$
    10. $(m_1^T - u_i m_3^T) \cdot P_i = 0, (m_2^T - v_i m_3^T) \cdot P = 0$
    11. further rearranging, we obtain
    12. $\begin{pmatrix} P_1^T & 0^T & -u_1 P_1^T \\ 0^T & P_1^T & -v_1 P_1^T \end{pmatrix} \begin{pmatrix} \vec{m}_1 \\ \vec{m}_2 \\ \vec{m}_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
    13. For n points, we can stack all these equations into a big matrix:
    14. $\begin{pmatrix} P_1^T & 0^T & -u_1 P_1^T \\ 0^T & P_1^T & -v_1 P_1^T \\ & \vdots & \\ P_n^T & 0^T & -u_n P_n^T \\ 0^T & P_n^T & -v_n P_n^T \end{pmatrix} \begin{pmatrix} \vec{m}_1 \\ \vec{m}_2 \\ \vec{m}_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}$
    15. $QM = 0$, Q (this matrix is known), M is the unknown matrix.
    16. Minimal solution
        1. $Q_{2n \times 12}$ should have rank 11 to have a unique (up to scale) non-zero solution M.

2. Because each 3D-to-2D point correspondence provides 2 independent equations, then $5 + \frac{1}{2}$ point correspondences are needed (in practice **6 point** correspondences)

17. Over-determined solution
    1. In theory we only need at least 5.5 point correspondences (each correspondences gives us 2 independent equations x and y) however in practice there is always noise, so we need to use overdetermined solution.
    2. For $n \geq 6$ points, a solution is the **Least squares solution**, which minimizes the sum of squared residuals, $||QM||^2$, subject to the constraint $||M||^2 = 1$. It can be solved through SVD.
    3. The solution is the eigenvector corresponding to the smallest eigenvalue of the matrix Q^TQ (because it is the unit vector x that minimizes $||Qx||^2 = x^TQ^TQx$).
    4. Matlab instructions:
    5.
    ```
    [U, S, V] = SVD(Q);
    M = V(:, 12);
    ```
    6. QR decomposition decomposes a matrix into orthogonal and upper triangular matrix.
    7. Once we have determined M, we can recover the intrinsic and extrinsic parameters by remembering that:
    8. $M = K(R|T)$
    9. Recommendation: use many more than 6 points (ideally more than 20) and non coplanar.
18. Reprojection error:
    1. The reprojection error is the Euclidean distance (in pixels) between an observed image point and the corresponding 3D point reprojected onto the camera frame.
    2. The reprojection error gives us a quantitative measure of the accuracy of the calibration (ideally it should be zero).
    3. $\pi(P_W^i, K, R, T)$ Reprojected point, $||p^i - \pi(P_W^i, K, R, T)||^2$
    4. The reprojection error can be used to assess the quality of the camera calibration
    5. what are the sources of the reprojection error? -> limitation of DLT
        1. Lens distortion
        2. Errors in corner detections
        3. Least square optimization is prone to noise
19. Non-linear calibration refinement
    1. The calibration parameters K, R, T determined by the DLT can be refined by minimizing the following cost: $K, R, T, \text{lens distortion} = argmin_{K,R,T,lens} \sum_{i=1}^{n} ||p^i - \pi(P_W^i, K, R, T)||^2$
    2. This time we also include the lens distortion (can be set to 0 for initialization).
    3. Can be minimized using Levenberg-Marquardt (more robust than Gauss-Newton to local minima).
20. Summary:
    1. M has 5 intrinsic parameters, 6 extrinsic parameter, $5 + 6 = 11$. M is a 3×4 matrix.
    2. Need at least 6 points to be able to estimate all the 12 unknowns.
    3. Note: each pair of point correspondences give rise to **two equations** since $\vec{x} = M\vec{X}$, is a homogenous equation, with x of size 3×1, M of size 3×4 and $\vec{X}$ of size 4×1, $x_1 = \frac{P_1^T\vec{x}}{P_3^T\vec{x}}$ and $x_2 = \frac{P_2^T\vec{x}}{P_3^T\vec{x}}$. Therefore, each point correspondences gives two constraints and we need at least 12 constraints to solve for the 12 unknowns, which means at least 6 points with known positions in 3D.
    4. $Mp = 0$, can be solved using SVD.

5. Camera calibration vs Triangulation
    1. Both are for solving linear equations
    2. Camera calibration: Getting camera parameters from
        1. estimated 2D matching
        2. 3D measurements
    3. Triangulation: Getting 3D points from
        1. estimated 2D matching
        2. camera parameters
6. Camera localization
    1. The extrinsic camera parameter aka. camera pose is not constant unless the camera is placed fixed to a world coordinate frame.
    2. PnP is used to get the transformation from world to camera given the point correspondences of 3D-2D. Or more formally:

3. Camera pose estimation: Given a set of 3D points wrt. a world coordinate system and their projecting points in the image plane, compute the rigid transformation between the world and the camera coordinate systems.
4. For perspective imaging device, this is the so-called *perspective n point (PnP)* problem.
5. PnP problem: n 3D-2D correspondences (i.e. correspondences between 3D model points and image points) $\rightarrow$ Find Rotation and Translation (R, T).
6. EPnP: An accurate $O(n)$ solution to the PnP problem.

References:

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